Program: Traffic Shaping using Leaky Bucket Algorithm

```c
#include <stdio.h>

int main() {
    int bucket_size, output_rate, n;
    int packet_size, stored = 0;

    printf("Enter bucket size: ");
    scanf("%d", &bucket_size);
    printf("Enter output rate: ");
    scanf("%d", &output_rate);
    printf("Enter number of packets: ");
    scanf("%d", &n);

    for (int i = 1; i <= n; i++) {
        printf("\nEnter packet size %d: ", i);
        scanf("%d", &packet_size);

        if (stored + packet_size <= bucket_size) {
            stored += packet_size;
            printf("Packet accepted.\n");
        } else {
            printf("Packet dropped due to buffer overflow.\n");
        }

        if (stored > 0) {
            int sent = (stored < output_rate) ? stored : output_rate;
            stored -= sent;
            printf("Packets sent: %d\n", sent);
            printf("Remaining in bucket: %d\n", stored);
        }
    }

    while (stored > 0) {
        int sent = (stored < output_rate) ? stored : output_rate;
        stored -= sent;
        printf("\nPackets sent: %d", sent);
        printf("\nRemaining in bucket: %d\n", stored);
    }
    return 0;
}
```

Output:

```
Enter bucket size: 10
Enter output rate: 4
Enter number of packets: 3

Enter packet size 1: 5
Packet accepted.
Packets sent: 4
Remaining in bucket: 1

Enter packet size 2: 6
Packet accepted.
Packets sent: 4
Remaining in bucket: 3

Enter packet size 3: 9
Packet dropped due to buffer overflow.
Packets sent: 3
Remaining in bucket: 0
```